

Smart City Bike Sharing Analysis

Objective:

This project analyzes smart bike-sharing station data using Power BI to understand bike availability, station performance, utilization efficiency, and city-wise operational patterns.

Problem Statement:

- Evaluate the overall utilization of bikes and stations to assess system efficiency
- Identify availability patterns across cities and stations (high, limited, and no availability)
- Analyze trends and detect underperforming areas to improve bike distribution and operational performance

Attributes:

Station ID/Name, City (Contract Name), Address, Latitude, Longitude, Date, Year, Month, Total Bikes, Bike Stands, Available Bikes, Empty Stands, Active Stations, Availability Status, Utilization %, Utilization Category

Data Preparation:

Performed data cleaning (removed duplicates, handled nulls, fixed address inconsistencies), created calculated columns (imputed address, latitude & longitude, date-only, availability status, utilization category), and developed DAX measures with a measure table along with a calendar table for time-based analysis.

Data Modelling:

The model uses a **star schema** with the Calendar Table as a time dimension, connected to BikeStationDB in a **one-to-many relationship** (one date → many station records).

Insights:

- **Capacity imbalance exists**, with most stations having **20–40 stands** while only a few large hubs reach **70 stands**, indicating uneven infrastructure distribution
- **Bike availability is highly concentrated** in major cities like **Bruxelles (3,532)**, **Lyon (3,475)**, and **Toulouse (2,686)**, whereas cities such as **Rouen (1)** and **Vilnius (0)** show negligible availability
- **System shows significant availability issues**, with **726 stations under limited availability** and **470 stations completely empty**, compared to only **150 well-performing stations**
- **Rapid growth in bike fleet** from **50 (2022)** to **19,450 (2025)** indicates expansion, but distribution has not scaled evenly
- **Frequent zero-bike situations at station level** highlight poor redistribution, while a maximum of **28 bikes at certain stations** indicates localized oversupply
- **High-demand cities are under pressure**, with **Toulouse (334)**, **Bruxelles (317)**, and **Lyon (292)** having many stations in limited availability, while high availability remains minimal (e.g., **Lyon 57**, **Valence 30**)
- The overall utilization rate is **34%**, showing that many stations are underutilized.

Conclusion:

The bike-sharing system is expanding rapidly; however, **bike distribution and station utilization remain highly uneven**. Many stations face shortages while demand is concentrated in major cities. This highlights the need for **better resource allocation and redistribution strategies** to ensure balanced availability and improved system efficiency.

Dashboard:

